

Lecture 1.2 – How Computers Work

Specific Learning Objectives:

1.1.8 – Understand the concept of working directories.

2.1.1 – Understand how file systems are structured and organized.

2.1.2 – Understand how to navigate file systems using a GUI interface.

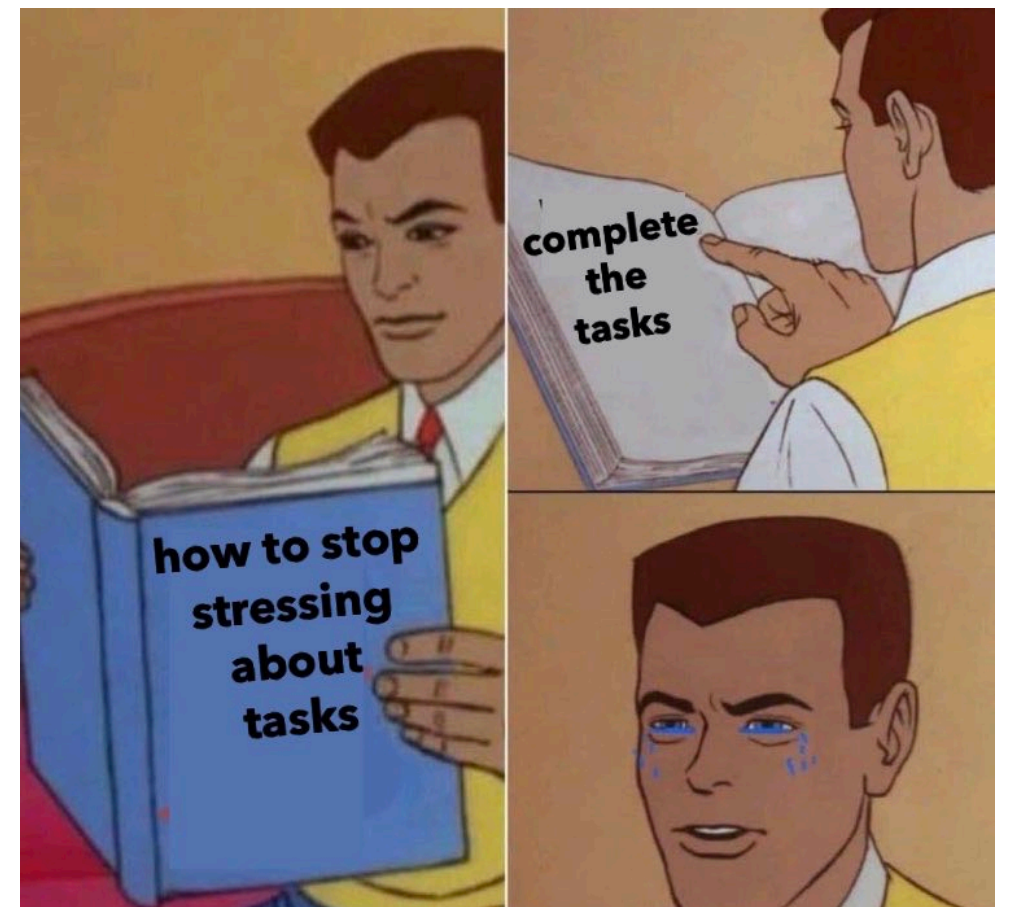
2.1.3 – Use directories to organize course work.

DON'T WORK IN YOUR DOWNLOADS FOLDER.



Tips for Excelling in this Course

- **Give yourself time to work (and put it down).** One of the most important tips is to schedule time to work on course content. Like, specific times in your calendar to work on coding assignments. **Don't get behind, and if you do, contact me for help!**
- **Work together!** There is nothing wrong with completing assignments together, just put your companions' names down on your assignment. Mix working together with working individually so that you can gain independence.
- **Coding is not as scary as you think.** Everyone in here can do this. But it's sometimes hard to fight the feeling. Be sure you're practicing recognizing and managing your own feelings about the course content. Why do you feel nervous? What could help you feel more comfortable? Actively seek these things out! **Do not respond by avoidance and procrastination, this will make the problem worse!**



Tips for Excelling in this Course

- **Use the internet.** Seriously, just google it first. Just about every question ever asked is only Stack Overflow or some similar website. If you pull a bit of code from somewhere, just include the URL in a comment on the assignment!
- **How is Google different than generate AI?** It takes a little more work to search for and apply code from a site rather than asking Chat. This makes it more likely you'll learn from the experience!

When you can't find an example of R code to steal from  **stackoverflow**



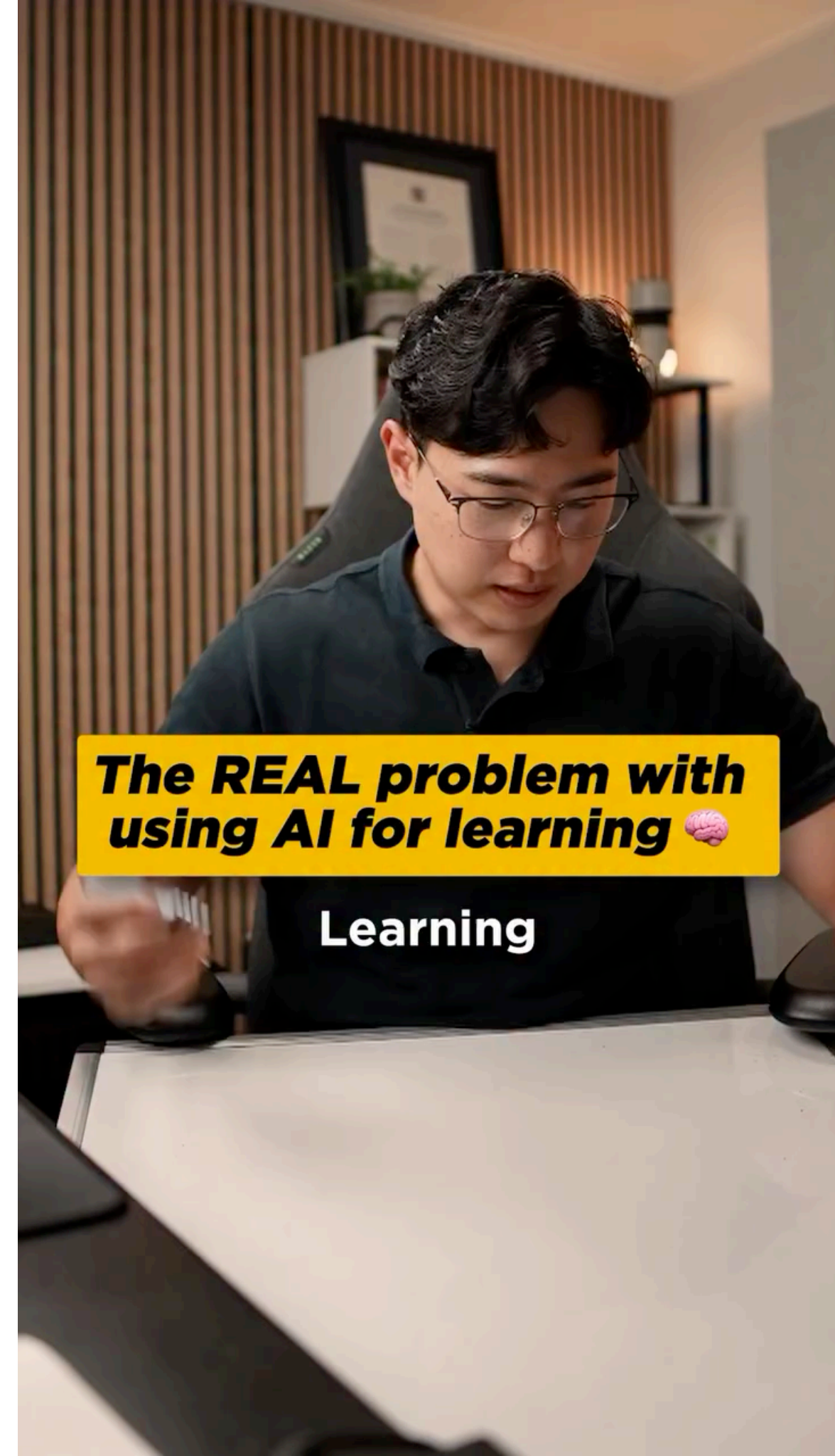
- **Just try stuff.** It's not 100% necessary that you understand exactly why something works for now. Just try stuff, and see if you can't narrow it down.

Why do I discourage the use of generative AI?

- Generative AI is a great way of bypassing the difficult process of learning.

Dr. Justin Sung on instagram

- Learning is hard, especially skills like problem solving and critical thinking. It's why it's such a valuable skill.
- Learning also sometimes feels bad because of confusion and frustration, but that means you're engaging thoughtfully (which is good).
- The **product** of a completed assignment is not nearly as important as the **process** of completing the work.
- Using an LLM to complete an assignment is robbing you of the process needed to learn!



The REAL problem with using AI for learning 🧠

Learning

Why do I discourage the use of generative AI?

- LLMs can sometimes explain things very well. But there is a huge difference between **understanding something** explained to you and **being able to do it**. I call this the “Bake-Off Effect”

Eli W McCann on Instagram

- Just watching the lectures without engaging in the material is just like watching Bake Off and thinking you can bake a cake!
- LLMs can magnify the effect via flattering and blandly positive language.
- Don't spend the next 4 weeks thinking you can make a mirror glaze cake and show up to the exam with a dud!



Before turning to generative AI...

- Before turning to generative AI, be sure to do these first:
 1. **Review the lecture slides and lecture notebook code.**
Most questions are derived from what's covered in lecture and require only small changes to existing code because they rely on the same conceptual basis.
 2. **Look in the textbook for examples or explanations.** The textbook has additional explanations and examples that may help you understand.
 3. **Search the internet with a standard search engine.** Even if you end up copying and pasting some code, try to understand why the code works. Some engagement is better than none!
 4. **Ask a knowledgeable person.** A classmate, student who has already taken the course, or me! We can help by explaining why the attempt isn't working, explain the solution, or give you next steps in figuring it out.

Using Generative AI in a way that might be helpful...

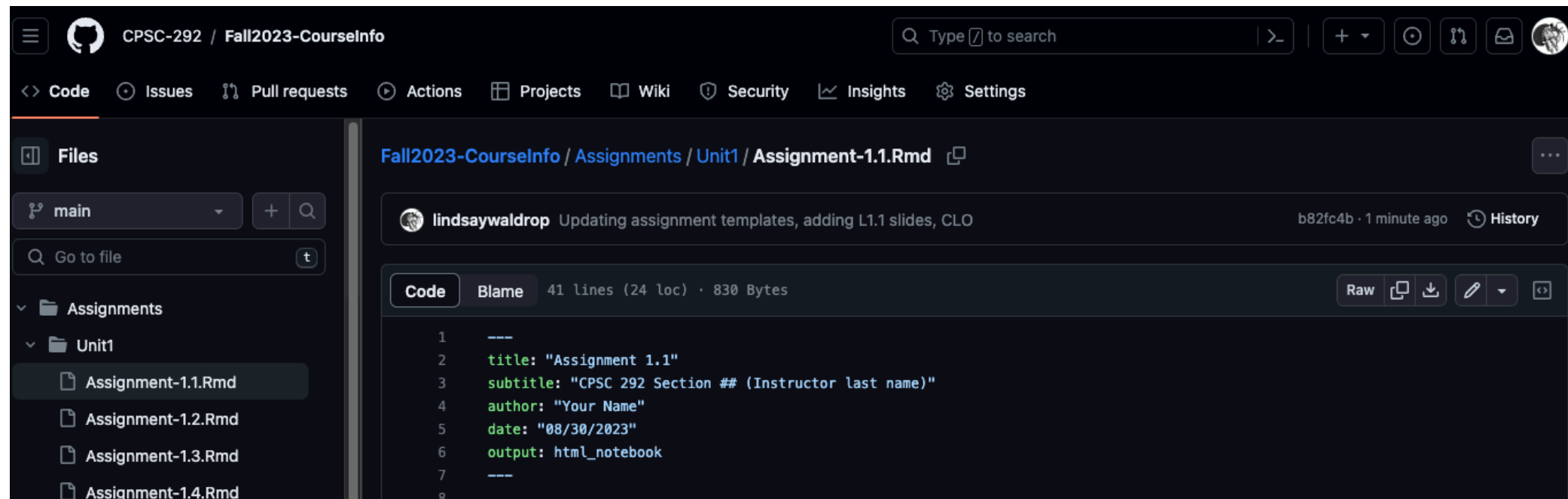
- If those don't work, using a Large Language Model could help:
 1. **Give it the problem and your attempt, have it explain why the attempt isn't working without giving you the answer.** Be sure to be specific with your prompt: don't give me the answer, explain to me why this doesn't work.
 2. **Prompt it to give you a clue without giving you the answer.** This is a very effective way of trying a problem without simply handing you the correct answer.
 3. **Ask the LLM to give you a similar problem to solve on your own.** If you end up having to ask for the solution and an explanation, don't stop there. Ask it to give you another, similar problem and be sure to work through it so you can cement the explanation and/or concept in your memory.

Remember: if you will not have the LLM during the exam. If you use the LLM in a way that avoids the work of learning, you will be disappointed with your performance on the exam.

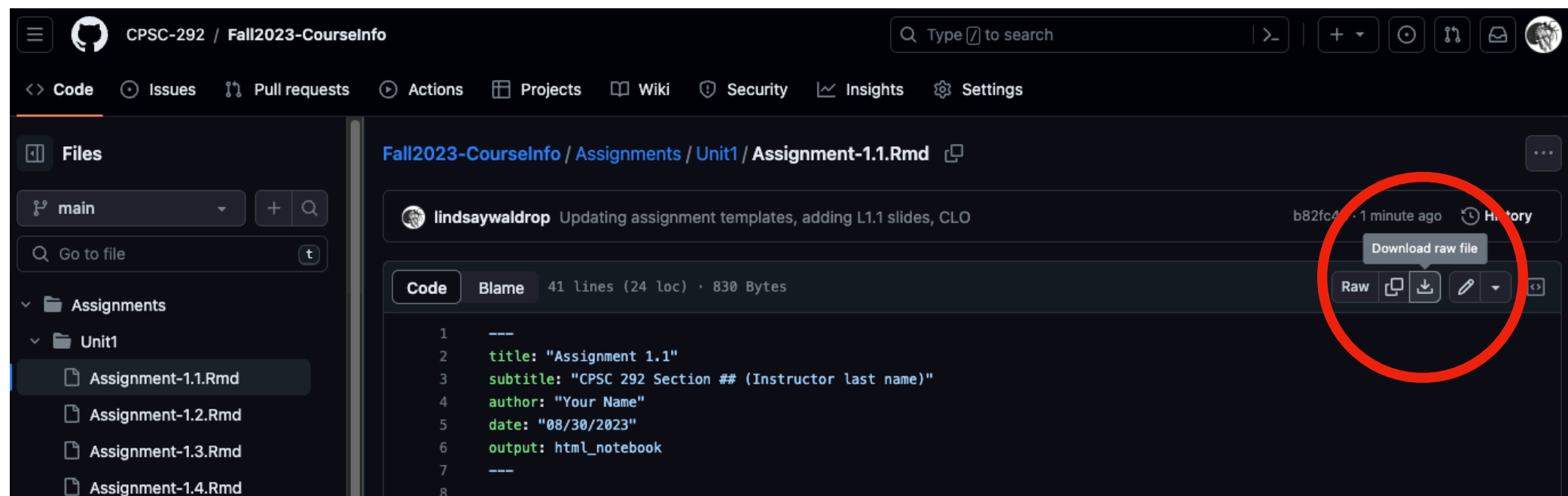
How to Download Assignment Templates

Method 1: Download raw file from Github.

1. Go to the course repository and navigate to the assignment.



2. Click on “download raw file”.

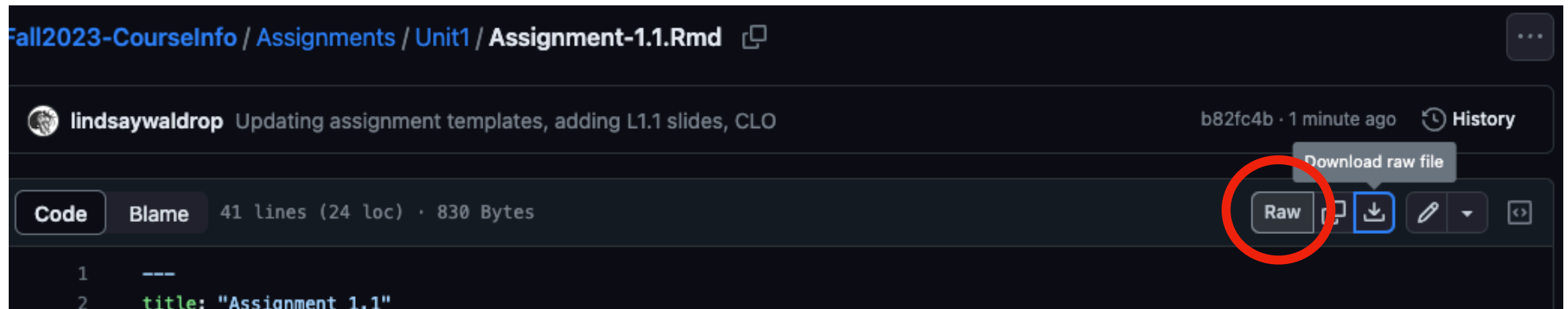


3. Move file OUT OF YOUR DOWNLOADS before working on it!

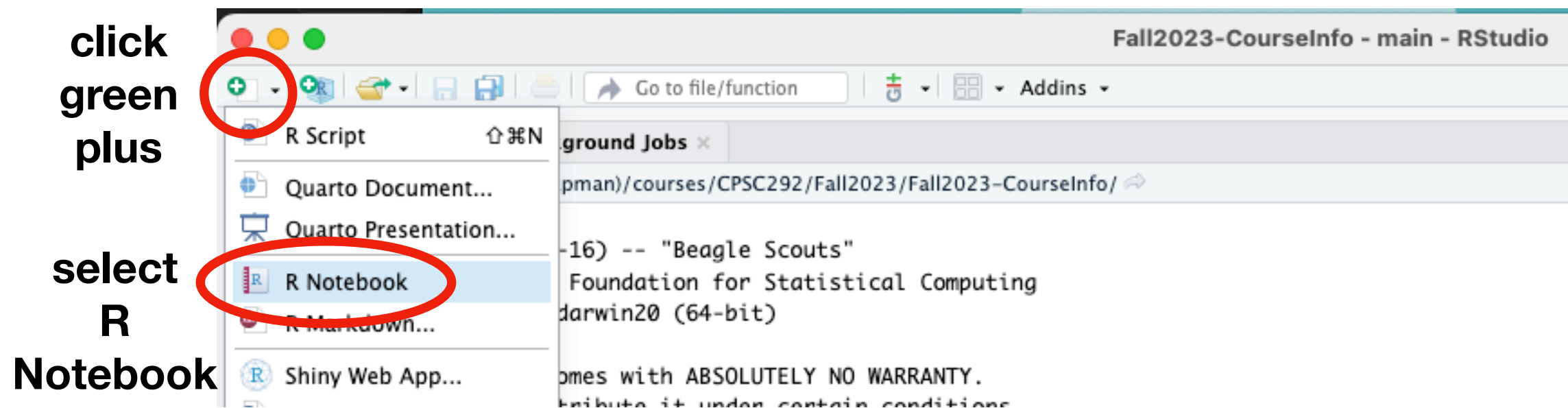
How to Download Assignment Templates

Method 2: Copy raw file from Github into R Notebook file.

1. Go to the course repository and navigate to the assignment.
2. Click on “raw” button. Highlight and copy ALL of the code on the page.



3. Open a new R notebook in RStudio.



4. REPLACE all the default text with the raw code from Github.
4. Save the file as the assignment name with extension .rmd

R Markdown Notebook Features

- R Markdown documents consist of **Markdown text** and **R code chunks**.

The image shows a screenshot of an R Markdown notebook editor window titled 'Untitled1'. The editor displays a sequence of alternating Markdown text and R code chunks. On the left side, three vertical brackets with labels identify these sections: 'Markdown text' for the first and third blocks, and 'R code "chunk"' for the second block. The first Markdown block (lines 1-8) includes a title 'R Notebook', output format 'html_notebook', and introductory text about R Markdown. The second R code chunk (lines 10-12) contains the R code `plot(cars)`. The third Markdown block (lines 14-18) provides instructions on adding new chunks and previewing the document. The editor's toolbar at the top includes buttons for navigation, saving, previewing, and running code. The text 'Writing goes here.' appears twice, once in each Markdown block, indicating where to write content. The text 'R code goes here!' appears in the R code chunk, indicating where to write code.

Markdown text

```
1 ---
2 title: "R Notebook"
3 output: html_notebook
4 ---
5
6 This is an [R Markdown](http://rmarkdown.rstudio.com) Notebook. When you execute code within the
7 notebook, the results appear beneath the code.
8 Try executing this chunk by clicking the *Run* button within the chunk or by placing your cursor
9 inside it and pressing *Cmd+Shift+Enter*.
```

R code "chunk"

```
10 ```{r}
11 plot(cars)
12 ```
```

Markdown text

```
13
14 Add a new chunk by clicking the *Insert Chunk* button on the toolbar or by pressing
15 *Cmd+Option+I*.
16 When you save the notebook, an HTML file containing the code and output will be saved alongside it
17 (click the *Preview* button or press *Cmd+Shift+K* to preview the HTML file).
18 The preview shows you a rendered HTML copy of the contents of the editor. Consequently, unlike
19 *Knit*, *Preview* does not run any R code chunks. Instead, the output of the chunk when it was
20 last run in the editor is displayed.
```

- Right now, we'll only use Markdown text. We'll start R code next week!

Today's Activity: Start Assignments 1.1, 1.2, & 1.3

- **We'll have “stations” for each assignment and rotate.** This is so everyone can start each assignment, get help, clarification, etc. I'll time you, and tell you when to “rotate.”
- **Work with different people at each “station.”** Try to switch up the people you work with so that you get a chance to meet new people.
- **Start by downloading the RMD file.** Open it up in RStudio and just try to get it to knit or preview.

Action Items

- 1. Complete assignments 1.1, 1.2, and 1.3 (due a week from today).**
- 2. No video for Friday's class. It will be a device-free day!**